

a) What is transient and resident virus? Explain trap door and logic bomb.

Transient vs. Resident Virus

Viruses are broadly categorized based on how they operate in memory:

- **Transient Virus:** This is a "non-resident" virus. Its lifecycle is tied directly to the host program it has infected.
 - It becomes active **only when the user runs the infected program**.
 - Once active, it executes its logic (e.g., finding other programs to infect) and then passes control back to the host program, which continues to run normally.
 - The virus is **not left active in the system's memory** after the host program is closed. It must wait to be executed again.
- **Resident Virus:** This is a more persistent and dangerous type of virus.
 - When an infected program is run, the resident virus **loads itself into the computer's main system memory (RAM)**.
 - It then "hooks" into the system's operations, such as file access or API calls.¹
 - Even after the original infected program is closed, the virus **remains active and "resident" in memory**.² From this position, it can monitor system activity and infect any new program that is run or any file that is accessed, making it spread much more effectively.³

Trap Door (or Backdoor)

A **trap door**, more commonly known as a **backdoor**, is a secret, undocumented method of bypassing normal authentication or security controls to gain access to a computer system, application, or data.⁴

- **How it's created:**
 - **By Developers:** Sometimes, programmers intentionally build in a special username/password or access mechanism to help with testing and debugging. If this "maintenance hook" is accidentally left in the final product, it becomes a major vulnerability.
 - **By Attackers:** An attacker who gains initial access to a system will often install a backdoor (like a remote access tool) so they can easily get back in later, even if their original entry point (like a stolen password) is discovered and fixed.

Logic Bomb

A **logic bomb** is a piece of malicious code intentionally inserted into a software system that will execute a malicious function only when one or more specific conditions are met.⁵

- **The Trigger:** The key feature is that it lies **dormant and undetected** until its "trigger" condition is satisfied. This trigger could be:
 - **Date/Time:** The most common type, known as a "time bomb," is set to go off on a specific date or time (e.g., Friday the 13th, or an employee's last day).⁶
 - **User Action:** The presence or absence of a specific file (e.g., a "disgruntled" employee's code might trigger if their name is removed from the employee database).
 - **System State:** A certain value in a database or reaching a specific number of transactions.
- **The Payload:** Once triggered, the logic bomb delivers its "payload," which could be anything from deleting data and corrupting files to crashing the entire system.⁷

b) What is Buffer Overflow attack and how is it prevented?

A **buffer** is a temporary data storage area in a computer's memory (RAM) that has a fixed, predefined size.⁸ For example, a program might set aside a buffer of 1,024 bytes to hold a user's name.

A **buffer overflow** is a software vulnerability that occurs when a program tries to write **more data** into a buffer than it was designed to hold.⁹ The excess data "overflows" the buffer's boundary and overwrites adjacent memory locations.¹⁰

The Attack:

An attacker exploits this by sending carefully crafted input that is intentionally too large. This input contains a "payload" of malicious code.

1. **Overflow:** The attacker's input overflows the buffer.
2. **Overwrite:** The excess data overwrites critical control data in memory, most notably the **return address**. The return address tells the program where to go next after it finishes its current task.¹¹
3. **Hijack:** The attacker overwrites the return address with the memory address of their own malicious code (called "shellcode") that they've just injected.
4. **Execute:** When the program finishes its task, it "returns" to the new, malicious address

and begins executing the attacker's code, giving the attacker control over the program and potentially the entire system.

How is it Prevented?

- **Secure Coding Practices:**
 - **Use "Safe" Functions:** Programmers (especially in languages like C/C++) must avoid dangerous, unbounded functions like `gets()` or `strcpy()`.¹² Instead, they should use "safe" alternatives like `fgets()` or `strncpy()`, which require specifying a maximum size to prevent overruns.
 - **Bounds Checking:** Actively checking the size of all input *before* writing it to a buffer.
- **Compiler-Level Protections:**
 - **Stack Canaries:** The compiler inserts a secret, random value (a "canary") in memory just before the return address. If a buffer overflow occurs, this canary value will be overwritten. The program checks the canary's value before returning; if it has changed, the program detects the attack and shuts down safely.¹³
- **Operating System-Level Protections:**
 - **Address Space Layout Randomization (ASLR):** This technique randomizes the memory addresses where program code and data are stored each time the program runs.¹⁴ This makes it extremely difficult for an attacker to predict the exact address of their malicious code, which they need to know to hijack the program's execution.
 - **Data Execution Prevention (DEP):** This marks certain areas of memory (like the stack, where buffers are often stored) as **non-executable**.¹⁵ Even if an attacker successfully injects their malicious code, DEP prevents the computer from running it.¹⁶

c) How is VPN helpful in securing data?

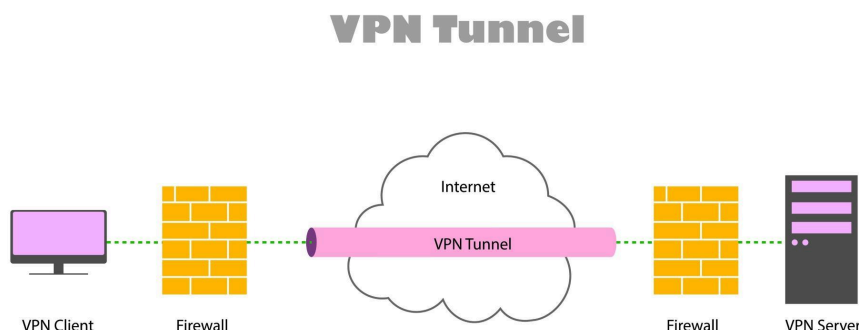
A **Virtual Private Network (VPN)** secures data by creating a private, encrypted connection for your device over a public network (like the internet).¹⁷ It does this through two primary mechanisms:

1. **Encryption (The "Tunnel"):**

- A VPN client on your device establishes an encrypted "tunnel" to a VPN server.¹⁸
- All your internet traffic—every website you visit, every file you send, every password you type—is first encrypted *before* it leaves your device.¹⁹
- This encrypted data travels through the tunnel to the VPN server, which then decrypts it and sends it to its final destination (e.g., the website you are visiting).²⁰
- **Security Benefit:** Anyone trying to "eavesdrop" on your connection—such as an attacker on a public Wi-Fi network, your Internet Service Provider (ISP), or a network administrator—cannot read your data.²¹ They can only see encrypted, unreadable gibberish.

2. **IP Address Masking (Anonymity):**

- Without a VPN, every website you visit sees your real, unique IP address, which reveals your general location and can be used to track you.
- When you use a VPN, all your traffic appears to come from the **VPN server's IP address**, not your own.²²
- **Security Benefit:** This hides your real IP address and location from websites and online services, protecting your privacy and making it much harder for trackers to build a profile of your online activities.



d) Countermeasures for Cloud Security

Securing the cloud is a "shared responsibility" between the cloud provider (like AWS, Google Cloud, Azure) and the customer.²⁴ The provider secures the *infrastructure*, while the customer must secure their *data and applications* running on that infrastructure.

Key countermeasures include:

- **Identity and Access Management (IAM):**
 - **Multi-Factor Authentication (MFA):** The single most effective countermeasure.²⁵ Requires users to provide two or more verification factors to gain access.
 - **Principle of Least Privilege:** Giving each user or service *only* the minimum permissions necessary to perform its job, and nothing more.
 - **Role-Based Access Control (RBAC):** Assigning permissions to roles (e.g., "developer," "auditor") rather than individual users, which is easier to manage.²⁶
- **Data Encryption:**
 - **Encryption in Transit:** Using protocols like TLS (HTTPS) to encrypt all data as it moves between users and the cloud, or between cloud services.²⁷
 - **Encryption at Rest:** Encrypting all data that is stored on disks or in databases, so it's unreadable if the physical storage is ever compromised.
- **Network Security:**
 - **Virtual Private Cloud (VPC):** Creating an isolated, private section of the cloud where you can launch resources.²⁸
 - **Security Groups & Firewalls:** Acting as virtual firewalls to control all incoming and outgoing traffic to your resources (e.g., "only allow web traffic on port 443").²⁹
 - **Network Segmentation:** Separating networks into smaller, isolated subnets (e.g., keeping the database servers on a separate network from the web servers).³⁰
- **Monitoring, Logging, and Auditing:**
 - Continuously logging all API calls and network traffic.
 - Using automated tools (like SIEM - Security Information and Event Management) to monitor these logs in real-time for suspicious activity and generate alerts.

- **Secure Configuration:**

- One of the most common cloud breaches is a simple **misconfiguration**.³² This includes things like leaving a storage bucket (like an S3 bucket) open to the public internet.
- Countermeasures involve using "Infrastructure as Code" (IaC) to automate and validate configurations and running regular audits for misconfigurations.

e) Different Types of Mobile Security Threats

Mobile devices (smartphones and tablets) face a unique set of threats:

- **Malicious Applications (Malware):**

- Apps, often downloaded from unofficial "third-party" app stores, that contain hidden malware.³³ This can include:
 - **Spyware:** Secretly records your calls, messages, location, and login credentials.
 - **Ransomware:** Encrypts all the data on your phone and demands a ransom payment to unlock it.³⁴
 - **Adware:** Aggressively displays unwanted ads.³⁵

- **Phishing (Smishing & Vishing):**

- These are attacks designed to steal your credentials (like banking passwords).³⁶
- **Phishing:** Deceptive emails.
- **Smishing:** Deceptive **SMS text messages** (e.g., "Your package delivery failed.³⁷ Click here to update your address.").
- **Vishing:** Deceptive **voice calls** or voicemails.

- **Insecure Network Connections:**

- **Unsecured Public Wi-Fi:** Many free Wi-Fi hotspots (in cafes, airports) do not encrypt traffic.³⁸ An attacker on the same network can easily intercept your data.
- **"Evil Twin" Attacks:** An attacker sets up a fake Wi-Fi network with a legitimate-sounding name (e.g., "Airport_Free_Wi-Fi").³⁹ When you connect, all your traffic goes through their computer, allowing them to steal passwords.

- **Data Leakage:**

- This is often unintentional. Poorly coded, legitimate apps may request more permissions than they need (e.g., a flashlight app asking for access to your contacts) and then accidentally leak or sell that sensitive data.⁴⁰
- **Outdated Operating System (OS):**
 - Many users ignore or delay OS updates. These updates often contain critical patches for known security vulnerabilities. Running an outdated OS is like leaving your front door unlocked.
- **Physical Theft or Loss:**
 - This is a simple but severe threat. If your device is lost or stolen and is not protected by a strong passcode, biometrics (face/fingerprint), and full-disk encryption, the thief gains access to all your personal data, photos, emails, and saved passwords.